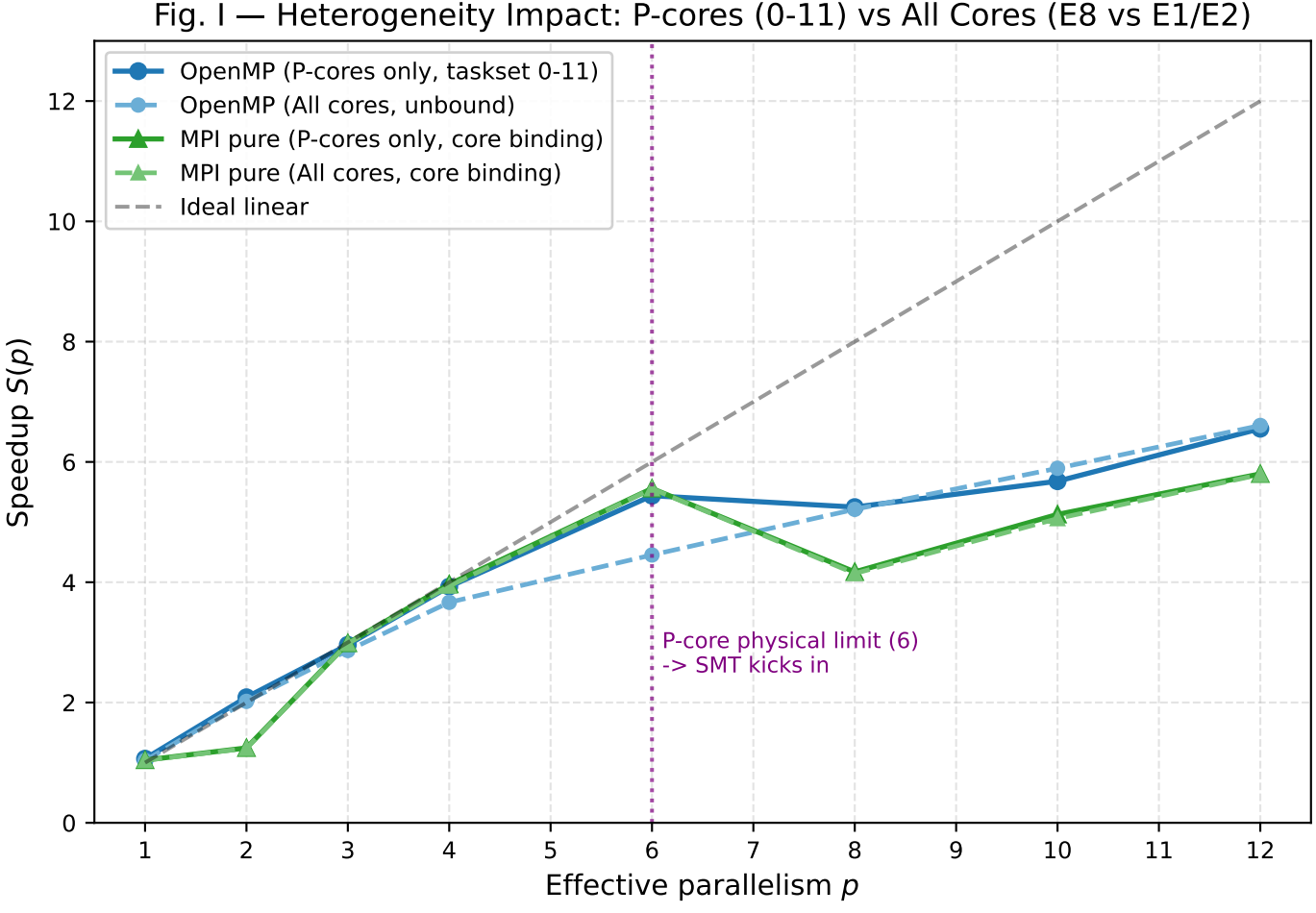
Fig. I — Heterogeneity Impact: P-cores (0-11) vs All Cores (E8 vs E1/E2)
OpenMP (P-cores only, taskset 0-11)
OpenMP (All cores, unbound)
MPI pure (P-cores only, core binding)
MPI pure (All cores, core binding)
Ideal linear
P-core physical limit (6)
-> SMT kicks in
Speedup S(p)
Effective parallelism p
0
2
4
6
8
10
12
1
2
3
4
5
6
7
8
9
10
11
12